A NON-GEODESIC EDGE THAT SURVIVES IN THE LOOP-ERASED ANT PROCESS

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Abstract. In the loop-erased ant process of Kious, Mailler and Schapira, ants walk one after another from a nest N to a food source F on a finite graph, each ant stepping along an edge with probability proportional to its weight and then adding one to the weight of every edge of the loop-erasure of its time-reversed trajectory. It was conjectured that the weight of an edge grows linearly exactly when the edge lies on a geodesic from N to F. Let G_ℓ be the graph made of a triangle N, u, v and two paths of ℓ edges, one from u to F and one from v to F. We prove that on G_ℓ the normalised weights converge almost surely. The limit is ρ on the edges of the path $N, u, \dots, F$, it is $1-\rho$ on the edges of the path $N, v, \dots, F$, and it is $(1-1/\ell)\rho(1-\rho)$ on the edge uv, where ρ is a non-degenerate random variable with values in $(0,1)$. For $\ell \geq 2$ the edge uv lies on no geodesic and keeps a positive share of the weight, so the conjecture fails on G_ℓ, and the proportion of ants that reinforce a path which is not a geodesic has a positive limit. For $\ell = 1$ the graph is the losange of Kious, Mailler and Schapira, the edge uv dies, and the theorem gives for the loop-erased rule the convergence that they proved for the uniform-geodesic rule. The forward-loop-erased ant process of Armfield and Mailler, in which the ant reinforces the loop-erasure of its trajectory instead of that of the reversed trajectory, has the same law, so all of these statements hold for it as well.

1. Introduction

1.1. **The process and the conjecture.** Let $G = (V, E)$ be a finite connected graph with two distinct marked vertices N (the nest) and F (the food). The *loop-erased ant process* of Kious, Mailler and Schapira [7] is a sequence of weight vectors $W(n) = (W_e(n))_{e \in E}$, $n \geq 0$, defined as follows. Initially $W_e(0) = 1$ for every edge. Given $W(n)$, the $(n+1)$-st ant performs a random walk $X = (X_0, \dots, X_K)$ on G started at $X_0 = N$ and stopped at the first time K at which it reaches F; from a vertex x it moves to a neighbour y with probability proportional to $W_{\{x,y\}}(n)$. Let $\bar{X}_j = X_{K-j}$ be the reversed trajectory, and let γ_{n+1} be its loop-erasure: the path $\gamma(0), \gamma(1), \dots$ with $\gamma(i) = \bar{X}_{j_i}$, where $j_0 = 0$ and $j_{i+1} = 1 + \max\{j : \bar{X}_j = \bar{X}_{j_i}\}$, stopped when it reaches N. This is a simple path from F to N, and

$$W_e(n+1) = W_e(n) + \mathbf{1}\{e \in \gamma_{n+1}\}, \qquad e \in E.$$

Conjecture 1 ([7, Conjecture 1.1(1)])**.** *For the loop-erased ant process on any finite graph, almost surely $W_e(n)/n \to \chi_e$ for all $e \in E$, where $\chi_e \neq 0$ almost surely if and only if the edge e belongs to at least one of the geodesics from N to F.*

Kious, Mailler and Schapira prove the conjecture for series-parallel graphs whose source and sink are N and F [7, Theorem 1.3]. For the simplest graph outside this class, the losange with vertices N, u, v, F and edges Nu, Nv, uv, uF, vF, they prove the corresponding statement for a second reinforcement rule, the uniform-geodesic rule [7, Theorem 1.4], and they write that for the loop-erased rule they could not prove convergence on the losange, although they believe that the same result holds. The general case is the first open problem of the survey [2], where the conjecture is stated in the form that the only edges whose weight grows linearly are

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those lying on a geodesic between N and F. A variant of the process with two nests is studied in [11]. The process in which every edge crossed by the walk is reinforced is studied in [8], where it is shown that it does not always find the geodesics. Other reinforced walks that model the formation of paths by ants are studied in [9, 5, 6].

Armfield and Mailler [3] consider the *forward-loop-erased* ant process, in which γ_{n+1} is the loop-erasure of the trajectory X itself and not of its reversal. They prove that if the distance from N to F is 2 then, whatever the rest of the graph, $W_e(n)/n \to 0$ almost surely for every edge e that lies on no geodesic, and they conjecture that this holds on every finite graph. The two processes have the same law: the loop-erasure of X is the first branch of Wilson's algorithm [13] rooted at F, so it is distributed as the path between N and F in the same random spanning tree as the loop-erasure of the reversed trajectory (Lemma 5 and the paragraph that follows its proof). The theorem of [3] therefore applies to the loop-erased process of [7], and the results of the present paper apply to the forward-loop-erased process.

1.2. **The result.** For $\ell \geq 1$ let G_ℓ be the graph with vertices N, u, v, F, $x_1, \dots, x_{\ell-1}$ and $y_1, \dots, y_{\ell-1}$, whose edges are Nu, Nv, uv, the edges of the path $u = x_0, x_1, \dots, x_\ell = F$ (the *u-leg*) and the edges of the path $v = y_0, y_1, \dots, y_\ell = F$ (the *v-leg*). We call uv the *rung*. The graph G_1 is the losange. The distance from N to F is $\ell + 1$, and there are two geodesics, $N, u, x_1, \dots, F$ and $N, v, y_1, \dots, F$. A path from N to F through the rung has $\ell + 2$ edges, so the rung lies on no geodesic.

Theorem 2. *Let $\ell \geq 1$ and consider the loop-erased ant process on G_ℓ. There is a random variable ρ with values in $(0, 1)$ such that, almost surely as $n \to \infty$,*

$$\frac{W_e(n)}{n} \longrightarrow \begin{cases} \rho & \text{if } e = Nu \text{ or } e \text{ is an edge of the u-leg,} \\ 1 - \rho & \text{if } e = Nv \text{ or } e \text{ is an edge of the v-leg,} \\ \left(1 - \frac{1}{\ell}\right)\rho(1-\rho) & \text{if } e = uv. \end{cases}$$

The variables ρ and $1 - \rho$ have the same law, and ρ is not almost surely constant.

Corollary 3. *Let $\ell \geq 2$. On G_ℓ the rung lies on no geodesic from N to F and $\chi_{uv} = (1-1/\ell)\rho(1-\rho) > 0$ almost surely, so Conjecture 1 fails on G_ℓ. Moreover the proportion of the first n ants whose reinforced path γ_k is not a geodesic converges almost surely to $(1-1/\ell)\rho(1-\rho)$, and*

$$\mathbb{P}(\gamma_n \text{ is not a geodesic}) \longrightarrow \left(1 - \tfrac{1}{\ell}\right) \mathbb{E}[\rho(1-\rho)] > 0.$$

Corollary 4. *For the loop-erased ant process on the losange, almost surely $W_e(n)/n \to \chi_e$ for every edge e, with $\chi_{Nu} = \chi_{uF} = 1 - \chi_{Nv} = 1 - \chi_{vF} \in (0, 1)$ and $\chi_{uv} = 0$.*

Corollary 4 is the case $\ell = 1$ of Theorem 2; it is the statement of [7, Theorem 1.4] with the loop-erased rule in place of the uniform-geodesic rule, and it agrees with Conjecture 1. The smallest graph of the family on which the conjecture fails is G_2, which has six vertices and seven edges. The half of the conjecture which says that the edges of the geodesics keep a positive share of the weight holds on every G_ℓ.

Since the two processes have the same law, Theorem 2 and its corollaries hold for the forward-loop-erased process of [3] as well. On the losange, where the distance from N to F is 2, the limit $\chi_{uv} = 0$ is a case of [3, Theorem 1.1]; the convergence of $W(n)/n$ under the assumptions of that theorem is stated in [3] as an open problem, and Corollary 4 gives it for the losange. In G_2 the distance from N to F is 3, so by Corollary 3 the conclusion of [3, Theorem 1.1] does not extend from distance 2 to distance 3.

The proof rests on two facts. The first is classical: the reinforced path of an ant has the law of the path from N to F in a random spanning tree of the weighted graph (Lemma 5). On G_ℓ

the spanning trees are easy to list, and this gives the law of γ_{n+1} given $W(n)$ in closed form (Lemma 6). The legs enter through the factor ℓ, the number of places where a leg can be cut: the total weight of the trees whose path from N to F uses the rung carries this factor, while the weight of the trees that contain both legs does not, and for $\ell \geq 2$ this is enough to keep the rung alive. The second fact is an exact martingale (Section 4.1), which gives the convergence of one coordinate of the process and from which the rest follows by elementary arguments.

2. The law of the reinforced path

Lemma 5. *Let G be a finite connected graph with positive edge weights (w_e) and let $N \neq F$ be two vertices. Let X be the weighted random walk started at N and stopped when it reaches F, and let γ be the loop-erasure of the reversed trajectory. Then γ has the law of the path between F and N in a random spanning tree T of G with $\mathbb{P}(T = t)$ proportional to $\prod_{e\in t} w_e$.*

Proof. For a vertex $x \neq N$ visited by the walk let σ_x be the first time at which the walk is at x, and let $p(x) = X_{\sigma_x - 1}$. In the reversed trajectory the last visit to a vertex x takes place at time $K - \sigma_x$, and the next vertex is $X_{\sigma_x-1} = p(x)$. By the definition of the loop-erasure, γ is therefore the path $F, p(F), p(p(F)), \dots$, followed until it reaches N. Now let the walk continue after time K until it has visited every vertex. By the theorem of Aldous and Broder [1, 4] (see also [10, Chapter 4]), the edges $\{x, p(x)\}$, $x \neq N$, form a spanning tree T with the law in the statement. The path from F to N in T is $F, p(F), p(p(F)), \dots$, because the first-visit times decrease along this sequence, and these vertices and their first-entrance edges are determined by the walk up to time K. Hence this path is γ. □

The loop-erasure γ' of X itself, in which the loops are removed in the order in which the walk closes them, has the same law as γ when both are regarded as sets of edges. Indeed, Wilson's algorithm [13] (see also [10, Chapter 4]) with root F and first starting vertex N produces a spanning tree with the law of Lemma 5, and the path between N and F in this tree is the loop-erasure of a weighted random walk started at N and stopped when it reaches F. The process in which the $(n+1)$-st ant adds one to the weight of every edge of the loop-erasure of its own trajectory is the forward-loop-erased ant process of [3]. It is therefore a Markov chain with the same initial state and the same transition probabilities as the loop-erased ant process, and every statement of this paper about $(W(n))_{n\geq 0}$ holds for it.

On G_ℓ there are four simple paths from N to F,

$$\pi_1 = (N, u, x_1, \dots, F), \qquad \pi_3 = (N, u, v, y_1, \dots, F),$$
$$\pi_2 = (N, v, y_1, \dots, F), \qquad \pi_4 = (N, v, u, x_1, \dots, F).$$

Each of them contains all the edges of one leg and no edge of the other. Hence in the ant process all the edges of the u-leg have the same weight at all times, and likewise for the v-leg. We write

$$A_n = W_{Nu}(n), \quad B_n = W_{Nv}(n), \quad R_n = W_{uv}(n),$$

and C_n, D_n for the common weight of the edges of the u-leg and of the v-leg. Since every π_i contains exactly one of Nu, Nv and exactly one of the two legs,

$$A_n + B_n = C_n + D_n = n + 2 =: S_n. \tag{1}$$

Lemma 6. *Give weight A to Nu, B to Nv, R to uv, C to each edge of the u-leg and D to each edge of the v-leg of G_ℓ. The path from N to F in the random spanning tree of Lemma 5 equals π_i with probability $q_i = m_i/(m_1 + m_2 + m_3 + m_4)$, where*

$$m_1 = AC(D + \ell B + \ell R), \qquad m_2 = BD(C + \ell A + \ell R), \qquad m_3 = \ell ARD, \qquad m_4 = \ell BRC,$$

and $m_1 + m_2 + m_3 + m_4 = CD(A + B) + \ell AB(C + D) + \ell R(A + B)(C + D)$.

Proof. The graph G_ℓ consists of three internally disjoint paths between u and v: the rung, the path u, N, v, and the path $u, x_1, \dots, F, \dots, y_1, v$ made of the two legs. It has $2\ell+2$ vertices and $2\ell+3$ edges, and its spanning trees are obtained by deleting one edge from each of two of the three paths; there are $2+6\ell$ of them. We list them with their weights divided by $C^{\ell-1}D^{\ell-1}$.

Delete the rung and Nu: the path is π_2 and the weight is BCD. Delete the rung and Nv: π_1, weight ACD. Delete the rung and one of the ℓ edges of the u-leg: π_2, weight ABD each. Delete the rung and an edge of the v-leg: π_1, weight ABC each. Delete Nu and an edge of the u-leg: π_2, weight BRD each. Delete Nu and an edge of the v-leg: π_4, weight BRC each. Delete Nv and an edge of the u-leg: π_3, weight ARD each. Delete Nv and an edge of the v-leg: π_1, weight ARC each. Summing over the trees with a given path gives $m_1, \dots, m_4$. □

3. The drift

Let $\mathcal{F}_n$ be the σ-field generated by the first n ants, and put

$$a_n = \frac{A_n}{S_n}, \quad b_n = \frac{B_n}{S_n} = 1 - a_n, \quad c_n = \frac{C_n}{S_n}, \quad d_n = \frac{D_n}{S_n} = 1 - c_n, \quad h_n = \frac{R_n}{S_n}.$$

By Lemmas 5 and 6 and by (1), $\mathbb{P}(\gamma_{n+1} = \pi_i \mid \mathcal{F}_n) = q_i(n)$ with

(2)

$$q_1 = \frac{ac(\ell b + d + \ell h)}{Q}, \quad q_2 = \frac{bd(\ell a + c + \ell h)}{Q}, \quad q_3 = \frac{\ell ahd}{Q}, \quad q_4 = \frac{\ell bhc}{Q}, \qquad Q = \ell ab + cd + \ell h,$$

where a, b, c, d, h, Q are evaluated at time n. Note that $0 < h_n < 1$ and $ab, cd \le \frac{1}{4}$, so

$$\ell\, a_n b_n + c_n d_n \le Q_n \le \tfrac{5\ell+1}{4}. \tag{3}$$

Lemma 7. *With $g = ad + bc - ab - cd/\ell$,*

$$\mathbb{P}(Nu \in \gamma_{n+1} \mid \mathcal{F}_n) = q_1 + q_3 = a + \frac{\ell ab(c-a)}{Q}, \qquad \mathbb{P}(Nv \in \gamma_{n+1} \mid \mathcal{F}_n) = b - \frac{\ell ab(c-a)}{Q},$$

$$\mathbb{P}(ux_1 \in \gamma_{n+1} \mid \mathcal{F}_n) = q_1 + q_4 = c + \frac{cd(a-c)}{Q}, \qquad \mathbb{P}(vy_1 \in \gamma_{n+1} \mid \mathcal{F}_n) = d - \frac{cd(a-c)}{Q},$$

$$\mathbb{P}(uv \in \gamma_{n+1} \mid \mathcal{F}_n) = q_3 + q_4 = h + \frac{\ell h(g-h)}{Q},$$

all quantities on the right being evaluated at time n.

Proof. Using $c+d=1$, $q_1+q_3-a = a\big(\ell bc + cd + \ell hc + \ell hd - \ell ab - cd - \ell h\big)/Q = \ell ab(c-a)/Q$. Using $a+b=1$, $q_1+q_4-c = c\big(\ell ab + ad + \ell ah + \ell bh - \ell ab - cd - \ell h\big)/Q = cd(a-c)/Q$. Finally $q_3+q_4-h = \ell h\big(ad + bc - ab - cd/\ell - h\big)/Q$. The two remaining identities follow because γ_{n+1} contains exactly one of Nu, Nv and exactly one of the two legs. □

When $a = c = p$ one has $g = (1-1/\ell)p(1-p)$. The points $a = c = p$, $h = (1-1/\ell)p(1-p)$, $0 < p < 1$, are thus the zeros of the drift with $h > 0$ when $\ell \ge 2$, and Theorem 2 says that the process converges to a random one of them. When $\ell = 1$ one has $g = (a-c)^2$.

4. Proof of Theorem 2

For an integer $k \ge 1$ let $\psi(k) = \sum_{j=1}^{k-1} 1/j$, so that $\psi(k+1) - \psi(k) = 1/k$. Comparing the sum with the integral of $1/x$,

$$0 \le \psi(k) - \log k \le 1, \qquad k \ge 1, \tag{4}$$

and $\psi(k) - \log k$ is non-decreasing in k and converges to Euler's constant. In particular, if (k_n) is a non-decreasing sequence of positive integers, then $\psi(k_n) - \log k_n$ converges. We also

use the following remark several times: if (k_n) is a non-decreasing sequence of positive integers with $k_{n+1} - k_n \in \{0, 1\}$, then

$$\sum_{n\ge 0} \frac{k_{n+1} - k_n}{k_n^2} \le \sum_{j\ge 1} \frac{1}{j^2} = \frac{\pi^2}{6}, \tag{5}$$

because each integer is the value of k_n at no more than one time n with $k_{n+1} = k_n + 1$.

4.1. **A martingale.** Let

$$M_n = \psi(A_n) - \psi(B_n) + \ell\big(\psi(C_n) - \psi(D_n)\big), \qquad r_n = \log \frac{A_n}{B_n} + \ell \log \frac{C_n}{D_n} = \operatorname{logit} a_n + \ell \operatorname{logit} c_n,$$

where $\operatorname{logit} x = \log(x/(1-x))$.

Lemma 8. *(M_n) is a martingale bounded in L^2. Consequently M_n converges almost surely and in L^2, r_n converges almost surely to a finite limit r_∞, and $r^* := \sup_n |r_n| < \infty$ almost surely.*

Proof. We have $\psi(A_{n+1}) - \psi(A_n) = \mathbf{1}\{Nu \in \gamma_{n+1}\}/A_n$, and similarly for B, C, D. By Lemma 7, with all quantities at time n,

$$\begin{aligned}
\mathbb{E}\big[\psi(A_{n+1}) - \psi(A_n) \mid \mathcal{F}_n\big] &= \frac{1}{aS}\Big(a + \frac{\ell ab(c-a)}{Q}\Big) = \frac{1}{S}\Big(1 + \frac{\ell b(c-a)}{Q}\Big),\\
\mathbb{E}\big[\psi(B_{n+1}) - \psi(B_n) \mid \mathcal{F}_n\big] &= \frac{1}{S}\Big(1 - \frac{\ell a(c-a)}{Q}\Big),\\
\mathbb{E}\big[\psi(C_{n+1}) - \psi(C_n) \mid \mathcal{F}_n\big] &= \frac{1}{S}\Big(1 + \frac{d(a-c)}{Q}\Big),\\
\mathbb{E}\big[\psi(D_{n+1}) - \psi(D_n) \mid \mathcal{F}_n\big] &= \frac{1}{S}\Big(1 - \frac{c(a-c)}{Q}\Big).
\end{aligned}$$

Hence

$$\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = \frac{1}{SQ}\Big(\ell(a+b)(c-a) + \ell(c+d)(a-c)\Big) = 0.$$

Write $M_{n+1} - M_n = U_n + \ell V_n$, where U_n equals $1/A_n$ or $-1/B_n$ according to whether γ_{n+1} contains Nu or Nv, and V_n equals $1/C_n$ or $-1/D_n$ according to the leg. By (5) applied to A, B, C and D, $\sum_n U_n^2 \le \pi^2/3$ and $\sum_n V_n^2 \le \pi^2/3$, so that $\sum_n (M_{n+1} - M_n)^2 \le 2(1+\ell^2)\pi^2/3$ surely. Since $M_0 = 0$, it follows that $\mathbb{E}[M_n^2] \le 2(1+\ell^2)\pi^2/3$ for all n, and M_n converges almost surely and in L^2.

Next, $M_n - r_n$ is the sum of the four terms $\pm(\psi(A_n) - \log A_n)$, …, $\pm\ell(\psi(D_n) - \log D_n)$. Each of the sequences A_n, B_n, C_n, D_n is non-decreasing, so each term converges, and by (4) $|M_n - r_n| \le 1 + \ell$. Therefore r_n converges almost surely to a finite limit and $\sup_n |r_n| < \infty$. □

4.2. **The process stays away from the boundary.**

Lemma 9. *Almost surely there is a (random) $\kappa_0 > 0$ such that $a_n, b_n, c_n, d_n \ge \kappa_0$ for all $n \ge 0$.*

Proof. Let $\kappa = (1 + e^{r^*/(1+\ell)})^{-1}$, so that $0 < \kappa \le \frac{1}{2}$. We first note that

$$a_n < \kappa \quad \Longrightarrow \quad a_n \le c_n. \tag{6}$$

Indeed, if $a_n > c_n$ then $\operatorname{logit} c_n < \operatorname{logit} a_n$, hence $-r^* \le r_n < (1+\ell)\operatorname{logit} a_n$, which gives $a_n > \kappa$.

Let $Y_n = \psi(A_n) - \log S_n$. By (4), $Y_n - 1 \le \log a_n \le Y_n$. Let

$$\Delta_n = \frac{\mathbf{1}\{Nu \in \gamma_{n+1}\}}{A_n} - \frac{1}{S_n}\Big(1 + \frac{\ell b_n(c_n - a_n)}{Q_n}\Big), \qquad \mathcal{N}_n = \sum_{k<n} \Delta_k.$$

By the first display in the proof of Lemma 8, $(\mathcal{N}_n)$ is a martingale. Its increments satisfy $\mathbb{E}[\Delta_n^2 \mid \mathcal{F}_n] \le \mathbb{E}[\mathbf{1}\{Nu \in \gamma_{n+1}\}/A_n^2 \mid \mathcal{F}_n]$, and by (5) the sum over n of the expectations of the right-hand side is at most $\pi^2/6$. So $\mathcal{N}_n$ is bounded in L^2, it converges almost surely, and $\mathcal{N}^* := \sup_n |\mathcal{N}_n| < \infty$ almost surely. Since $\log(1 + 1/S_n) \le 1/S_n$,

$$Y_{n+1} - Y_n = \frac{\mathbf{1}\{Nu \in \gamma_{n+1}\}}{A_n} - \log\Big(1 + \frac{1}{S_n}\Big) \ge \Delta_n + \frac{\ell b_n(c_n - a_n)}{Q_n S_n}. \tag{7}$$

Fix n with $a_n < \kappa$, and let m be the last time before n with $a_m \ge \kappa$; it exists because $a_0 = \frac{1}{2} \ge \kappa$. For $m < k < n$ we have $a_k < \kappa$, so $a_k \le c_k$ by (6) and the last term of (7) is non-negative at time k. Summing (7) over $m < k < n$ gives $Y_n \ge Y_{m+1} - 2\mathcal{N}^*$. Moreover $Y_{m+1} \ge Y_m - \log(1 + 1/S_m) \ge Y_m - \log\frac{3}{2}$ and $Y_m \ge \log a_m \ge \log\kappa$. Hence

$$\log a_n \ge Y_n - 1 \ge \log\kappa - \log\tfrac{3}{2} - 1 - 2\mathcal{N}^*.$$

The same bound holds trivially when $a_n \ge \kappa$. So $\inf_n a_n > 0$ almost surely.

The map exchanging u with v and x_i with y_i is an automorphism of G_ℓ that fixes N and F; it exchanges a_n with b_n and c_n with d_n. Hence $\inf_n b_n > 0$ almost surely as well. Finally $\ell \operatorname{logit} c_n = r_n - \operatorname{logit} a_n$ is bounded, so c_n and d_n are bounded away from 0. □

4.3. **The two geodesics.**

Lemma 10. *Almost surely $a_n - c_n \to 0$, and $a_n \to \rho$, $c_n \to \rho$, where $\rho = (1 + e^{-r_\infty/(1+\ell)})^{-1} \in (0, 1)$.*

Proof. Let $\delta_n = a_n - c_n$ and $\xi_{n+1} = \mathbf{1}\{Nu \in \gamma_{n+1}\} - \mathbf{1}\{ux_1 \in \gamma_{n+1}\}$. Since $a_{n+1} - a_n = (\mathbf{1}\{Nu \in \gamma_{n+1}\} - a_n)/(S_n + 1)$ and similarly for c_n,

$$\delta_{n+1} - \delta_n = \frac{\xi_{n+1} - \delta_n}{S_n + 1}, \qquad \mathbb{E}[\xi_{n+1} - \delta_n \mid \mathcal{F}_n] = -\lambda_n \delta_n, \qquad \lambda_n = \frac{\ell a_n b_n + c_n d_n}{Q_n},$$

by Lemma 7. As $|\xi_{n+1} - \delta_n| \le 2$,

$$\mathbb{E}[\delta_{n+1}^2 \mid \mathcal{F}_n] \le \delta_n^2 - \frac{2\lambda_n \delta_n^2}{S_n + 1} + \frac{4}{(S_n + 1)^2}.$$

By the theorem of Robbins and Siegmund [12], δ_n^2 converges almost surely and $\sum_n \lambda_n \delta_n^2/(S_n + 1) < \infty$ almost surely. By Lemma 9 and (3), $\lambda_n \ge 4(\ell + 1)\kappa_0^2/(5\ell + 1) > 0$ for all n, and $\sum_n 1/(S_n + 1) = \infty$; so the limit of δ_n^2 is 0.

The numbers a_n and c_n stay in $[\kappa_0, 1 - \kappa_0]$, on which logit is uniformly continuous; hence $\operatorname{logit} a_n - \operatorname{logit} c_n \to 0$, and with Lemma 8, $(1 + \ell)\operatorname{logit} a_n \to r_\infty$. This gives the limit of a_n, and $c_n = a_n - \delta_n$ has the same limit. □

4.4. **The rung.**

Lemma 11. *Let (β_n) be positive numbers with $\beta_n \to 0$ and $\sum_n \beta_n = \infty$, let (g_n) be real numbers converging to $g^* \ge 0$, and let (η_n) be a convergent real sequence. Let (u_n) satisfy*

$$u_{n+1} - u_n = \beta_n\big(g_n - e^{x_n}\big), \qquad x_n = u_n + \eta_n,$$

and assume that $|g_n - e^{x_n}| \le 2$ for all n. Then $e^{x_n} \to g^*$.

Proof. Fix $\theta > g^*$ and $\varepsilon > 0$, and choose n_1 such that for all $m, n \ge n_1$ we have $g_n \le (g^*+\theta)/2$, $2\beta_n \le \varepsilon$ and $|\eta_m - \eta_n| \le \varepsilon$. If $n \ge n_1$ and $x_n > \log\theta$, then $u_{n+1} - u_n \le -\beta_n(\theta - g^*)/2$. If this held for all large n, then $u_n \to -\infty$ and, η_n being bounded, $x_n \to -\infty$, which is absurd. So there is $k \ge n_1$ with $x_k \le \log\theta$. Let $n > k$ with $x_n > \log\theta$ and let m be the last time before n with $x_m \le \log\theta$, so that $m \ge k$. For $m < j < n$ we have $u_{j+1} \le u_j$, hence $u_n \le u_{m+1} \le u_m + 2\beta_m \le u_m + \varepsilon$ and $x_n \le x_m + 2\varepsilon \le \log\theta + 2\varepsilon$. Therefore $\limsup_n x_n \le \log\theta + 2\varepsilon$, and since $\theta > g^*$ and $\varepsilon > 0$ are arbitrary, $\limsup_n e^{x_n} \le g^*$. This proves the lemma when $g^* = 0$.

If $g^* > 0$, fix $0 < \theta < g^*$ and $\varepsilon > 0$, and choose n_1 such that for all $m, n \ge n_1$ we have $g_n \ge (g^*+\theta)/2$, $2\beta_n \le \varepsilon$ and $|\eta_m - \eta_n| \le \varepsilon$. If $n \ge n_1$ and $x_n < \log\theta$, then $u_{n+1} - u_n \ge \beta_n(g^* - \theta)/2$; this cannot hold for all large n, since then $x_n \to +\infty$. The same last-exit argument, with the inequalities reversed, gives $\liminf_n x_n \ge \log\theta - 2\varepsilon$, and so $\liminf_n e^{x_n} \ge g^*$. □

Lemma 12. *Almost surely* $h_n \to (1 - 1/\ell)\rho(1-\rho)$.

Proof. Let $g_n = a_n d_n + b_n c_n - a_n b_n - c_n d_n/\ell$. By Lemma 10, $g_n \to g^* := (1-1/\ell)\rho(1-\rho) \ge 0$. Let $Y'_n = \psi(R_n) - \log S_n$. By Lemma 7,

$$\mathbb{E}\big[\psi(R_{n+1}) - \psi(R_n) \mid \mathcal{F}_n\big] = \frac{1}{h_n S_n}\Big(h_n + \frac{\ell h_n(g_n - h_n)}{Q_n}\Big) = \frac{1}{S_n} + \beta_n(g_n - h_n), \qquad \beta_n = \frac{\ell}{Q_n S_n}.$$

Let $\Delta'_n = \mathbf{1}\{uv \in \gamma_{n+1}\}/R_n - 1/S_n - \beta_n(g_n - h_n)$ and $\mathcal{N}'_n = \sum_{k<n} \Delta'_k$. As in the proof of Lemma 9, $(\mathcal{N}'_n)$ is a martingale bounded in L^2, so it converges almost surely. With $e_n = 1/S_n - \log(1 + 1/S_n) \in [0, 1/(2S_n^2)]$ we get

$$Y'_{n+1} - Y'_n = \Delta'_n + \beta_n(g_n - h_n) + e_n.$$

Put $\varepsilon_n = \psi(R_n) - \log R_n$, which converges because R_n is non-decreasing, and

$$u_n = Y'_0 + \sum_{k<n} \beta_k(g_k - h_k), \qquad \eta_n = \mathcal{N}'_n + \sum_{k<n} e_k - \varepsilon_n.$$

Then η_n converges almost surely, $\log h_n = Y'_n - \varepsilon_n = u_n + \eta_n$, and $u_{n+1} - u_n = \beta_n(g_n - e^{u_n+\eta_n})$. We have $-\frac{1}{2} \le g_n \le 1$ and $0 < h_n < 1$, so $|g_n - h_n| \le 2$. By Lemma 9 and (3), $(\ell+1)\kappa_0^2 \le Q_n \le (5\ell+1)/4$, so $\beta_n \to 0$ and $\sum_n \beta_n = \infty$. Lemma 11 gives $h_n \to g^*$. □

4.5. **End of the proof.** Since $S_n/n \to 1$, Lemmas 10 and 12 give the limits in Theorem 2. The automorphism used in the proof of Lemma 9 exchanges ρ and $1-\rho$, so they have the same law. By Lemma 9 the weights A_n, B_n, C_n, D_n tend to infinity, so the four terms of $M_n - r_n$ in the proof of Lemma 8 tend to $\pm$ Euler's constant and $\pm\ell$ times Euler's constant and cancel; thus $r_\infty = \lim_n M_n$. As M_n converges in L^2, $\mathbb{E}[r_\infty] = 0$ and $\mathbb{E}[r_\infty^2] \ge \mathbb{E}[(M_1 - M_0)^2] > 0$, since $M_1 - M_0 = 1 + \ell$ when $\gamma_1 = \pi_1$. So r_∞, and with it ρ, is not almost surely constant. This proves Theorem 2.

Proof of Corollary 3. The paths π_3 and π_4 are the simple paths from N to F that are not geodesics, and they are the ones that contain the rung. So the number of $k \le n$ for which γ_k is not a geodesic is $R_n - 1$, and $(R_n - 1)/n \to (1-1/\ell)\rho(1-\rho)$ almost surely, which is positive when $\ell \ge 2$. By Lemma 7, $\mathbb{P}(\gamma_{n+1} \in \{\pi_3, \pi_4\} \mid \mathcal{F}_n) = h_n + \ell h_n(g_n - h_n)/Q_n$, which converges almost surely to the same limit because Q_n is bounded below; the last statement follows by bounded convergence. □

5. Simulations

The program `sim_ants.c` runs the process as defined in the introduction: it stores the walk of each ant, reverses it, erases the loops in chronological order and adds one to the weights of the edges of the resulting path. It makes no use of Lemma 5 or Lemma 6.

With the weights held fixed, the frequencies of the four paths agree with Lemma 6. For $\ell = 2$ and $(A, B, C, D, R) = (2, 3, 1, 4, 2)$, two million ants gave the frequencies 0.15544, 0.60037, 0.17762 and 0.06657 for $\pi_1, \dots, \pi_4$, and Lemma 6 gives 0.15556, 0.60000, 0.17778 and 0.06667. For $\ell = 3$ and the weights $(5, 2, 3, 4, 1)$ the frequencies were 0.44208, 0.38081, 0.13629 and 0.04083, and the lemma gives 0.44218, 0.38095, 0.13605 and 0.04082. The script `check_path_law.py` verifies Lemma 6 symbolically for $\ell \le 5$ by listing the spanning trees, and verifies the identities of Lemma 7.

Table 1 shows the state of the process after 10^8 ants in independent runs. In every run a_n and c_n differ by less than 0.0003, and for $\ell \ge 2$ the value of h_n is close to $(1 - 1/\ell)a_n(1 - a_n)$.

ℓ	seed	a_n	c_n	h_n	$(1 - 1/\ell)a_n(1 - a_n)$
2	1	0.38569	0.38574	0.12126	0.11847
2	2	0.94939	0.94938	0.02419	0.02402
2	3	0.14083	0.14079	0.06048	0.06050
2	4	0.31259	0.31262	0.11194	0.10744
2	5	0.53892	0.53897	0.12692	0.12424
2	6	0.12526	0.12523	0.05685	0.05478
3	1	0.64536	0.64537	0.15316	0.15258
3	2	0.93299	0.93305	0.04243	0.04168
5	1	0.53274	0.53295	0.19958	0.19914
5	2	0.41410	0.41408	0.19433	0.19410
10	1	0.35721	0.35711	0.20658	0.20665
1	1	0.46134	0.46132	0.02736	0
1	2	0.91163	0.91162	0.00766	0

TABLE 1. The loop-erased ant process on G_ℓ after $n = 10^8$ ants.

The convergence of h_n is slow when ℓ is small. Linearising the drift of h at the limit point gives the rate $-(1-1/\ell)/2$, which suggests that $h_n - (1-1/\ell)a_n(1-a_n)$ is of order $n^{-(1-1/\ell)/2}$, that is $n^{-1/4}$ for $\ell = 2$. In the run with $\ell = 2$ and seed 1 this difference was 0.0401, 0.0187, 0.0090, 0.0050, 0.0028 at $n = 10^4, \dots, 10^8$; the last ratios are 0.56, and $10^{-1/4} = 0.562$. For $\ell = 1$ the drift of h is $h(\delta^2 - h)/Q$ with $\delta = a - c$, so that $1/h_n$ should grow by about $\log 10/Q_n$ per decade of n. In the run with $\ell = 1$ and seed 1, h_n was 0.0519, 0.0423, 0.0357, 0.0311, 0.0274 at $n = 10^4, \dots, 10^8$, so $1/h_n$ grew by between 4.1 and 4.4 per decade, and $\log 10/Q_n$ is close to 4.4 there.

6. Remarks

Remark 13. On G_ℓ the limit of the rung is a function of the limit ρ of the two geodesics, and all edges of the geodesics survive. We do not know which edges survive on a general graph, nor whether the edges of the geodesics always do. By Lemma 5 the mean drift of the process on any graph is given by the laws of the paths between N and F in weighted random spanning trees, and on G_ℓ the rung survives because of the weight $CD(A + B)$ of the two trees that contain both legs, which is not multiplied by ℓ in Lemma 6: at $a = c = p$ and small h the rung is reinforced at $2\ell/(\ell + 1)$ times the rate h at which it is diluted, and this ratio exceeds 1 exactly when $\ell \ge 2$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [16] and Math Brain [14]. In October 2026 an AI agent connected to the Brain, working from the project's list of open problems in probability, found the graph G_2 and showed that on it the rung keeps a positive share of the weight with positive probability; a Codex agent (OpenAI) and a Claude agent (Anthropic) each checked that argument. In November 2026 a Claude Fable agent (Anthropic) found Theorem 2 and its proof, wrote the programs, ran the simulations and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/loop-erased-ants-non-geodesic-edge`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [14].

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